1-1 Information and Media

Point!

1 Data, Information, Knowledge

(1) ([1]**Data**) : Facts or matters represented using numbers, characters, or symbols.

(2) ([2]**Information**) : Something that holds meaning or value for the recipient and serves as a basis for decision-making. Unlike objects, information has no form.

- Information has the following characteristics.

[1] ([3]**Persistence**) : Information, once created, cannot be completely erased.

[2] ([4]**Reproducibility**) : Information can be easily copied in large quantities.

[3] ([5]**Propagation**) : Information is easy to convey and spread.
Information can easily spread through ([6]**mass media**) such as newspapers and television, as well as the internet.

(3) ([7]**Knowledge**) : Information that has been analyzed and systematized to aid in problem-solving.

(4) Primary information and secondary information

[1] ([8]**Primary information**) : Information obtained through direct personal experience or acquired through research and experiments.
<Examples> Experimental reports, survey results, tabulations of questionnaire results, etc.

[2] ([9]**Secondary information**) : Information not obtained directly by oneself, but acquired through a third party.
<Example> Books, newspapers, television, etc.

- Secondary information can sometimes differ from the original information or have interpretations added to it. In other words, it is necessary to compare the obtained information from multiple perspectives with other information to determine the degree of its accuracy and reliability. This is called ([10]**cross-checking**).

2 Media

(1) ([11]**Media**) : Media for conveying information to people.

(2) Types of media

Name	Content	Example
([12]**Expression media**)	Media used as a means of expressing information.	Text, images, audio, video
([13]**Propagation/ Transmission media**)	Media used as an intermediary for the transmission and communication of information.	Television, radio, newspapers, books, telephone, internet
([14]**Recording media**)	Media used for recording and storing information.	Paper, USB drives, DVDs, cloud storage

(3) ([15]**Media literacy**) : The ability to accurately interpret information obtained from media.

Warm Up

Answer the following questions.

(1) [1] For the following items a to c, answer whether they are related to: **1)** Data, **2)** Information, or **3)** Knowledge. Write "1," "2," or "3" as your answer.

a Temperature and humidity values

b Analysis results of temperature and humidity over the past 10 years

c Weather forecast

[2] Choose one statement that is not suitable as a characteristic of information from the options **A** to **D**, and answer using the letters.

A It has a form just like an object.

B The meaning or value changes depending on the recipient.

C It has the characteristic of persistence.

D It has the characteristic of propagation.

[3] Choose the content related to the following sentences a to c from the group of words labeled **A** to **C** below, and answer using the letters.

a Even if you purchase a commercially available music CD, you are not allowed to make copies to distribute to friends.

b Emails can be sent instantly overseas.

c Rumors about people and false information can persist indefinitely.

[Word group] **A** Persistence **B** Reproducibility **C** Propagation

(2) [1] For the following media from a to f, answer whether they are classified as **1)** Expression media, **2)** Propagation/Transmission media, or **3)** Recording media. Write "**1**," "**2**," or "**3**" as your answer.

a Television b Cloud storage c Photograph

d Internet e Text character f Book

[2] What is the term for the ability to accurately interpret information obtained from media?

Explanation

(1) [1] a: **1** b: **2** c: **3**

[2] Unlike objects, information does not have form. Therefore, **A**

[3] a: The characteristic of being easily copied is reproducibility. **B**

b: The characteristic of being able to be sent instantaneously is propagation. **C**

c: The characteristic of information, once created, that never disappears is persistence. **A**

(2) [1] a: **2** b: **3** c: **1** d: **2** e: **1** f: **2**

[2] Media literacy

Try

1 Answer the following questions.

(1) What is the term for the representation of facts or matters using numbers, characters, or symbols?

(2) For the following items a to c, answer whether they are related to: **1)** Data, **2)** Information, or **3)** Knowledge. Write "**1**," "**2**," or "**3**" as your answer.

a Results on mock exam b Analysis results for admission to the desired school c Scores on mock exam

(3) Choose the characteristic of information that is appropriate from the options **A** to **D** below, and answer using the letters.

A Information does not have the characteristic of reproducibility.
B The meaning or value of the information does not change depending on the recipient.
C Information has form like an object.
D Information that has been disseminated cannot be easily deleted.

(4) Choose the content related to the following sentences [1] to [3] from the group of words labeled **A** to **C** below, and answer using the letters.

[1] It is possible to copy commercially available music and movies in large quantities without any degradation.
[2] Rumors about people persist and never completely disappear.
[3] Information published on the internet spreads widely in a short period of time.
[Word group] **A** Persistence **B** Reproducibility **C** Propagation

(5) What is the term for information obtained through personal experience or from investigations and experiments that were personally conducted?

2 Answer the following questions.

(1) What is the term for the medium used to convey information to people?

(2) Choose all the media corresponding to the following a to c from the options **A** to **F**, and answer using the letters.

a Expression media b Propagation/Transmission media c Recording media
A USB drive **B** Radio **C** Telephone **D** DVD **E** Video **F** Audio

(3) Choose the explanation of media literacy that is appropriate from the options **A** to **D** below, and answer using the letters.

A The ability to accept information disseminated by the media without question
B The ability to understand the meaning and characteristics of media and to accurately interpret information
C The ability to capture content for media distribution
D The ability to use media to edit information in accordance with the intent of the sender

Exercise

1 Read the following passage and answer each question.

> We obtain (a) such as weather forecasts from sources such as newspapers and television, and use it as a basis for deciding on our actions. Weather forecasts organize (b) such as weather conditions, precipitation, and atmospheric pressure patterns, and add meaning and value to them. Additionally, this can be generalized to accumulate (c), such as "it often rains in May and June every year."

(1) For each of the following options **A** to **C**, use the corresponding letter to indicate which blank, (a) to (c), it is related to.

A Temperature graph **B** Temperature **C** Analysis results of the lowest temperatures over the past 20 years

(2) Choose the content related to the following sentences [1] to [3] from the group of words labeled **A** to **C** below, and answer using the letters.

[1] False information once disseminated on the internet can persist indefinitely.
[2] Not only mass media, but also individuals are now able to disseminate large amounts of information throughout the world.
[3] A person copied a commercially available music CD and distributed it to their friends.

A Persistence **B** Reproducibility **C** Propagation

(3) For the following statements about information, mark "○" if the statement is correct, and "×" if it is incorrect.

[1] Primary information refers to information obtained through surveys or investigations conducted by the person themselves.
[2] The statistical data published on the Statistics Bureau's website is considered primary information.
[3] Information found by a person on a website is considered primary information.
[4] Information obtained through a third party is considered secondary information.

2 Answer the following questions.

(1) Choose all of the following media from **A** to **F** that qualify as Propagation/Transmission media.

A Internet **B** Text character **C** USB drive
D Audio **E** Radio **F** Book

(2) What is the term for the ability to understand the meaning and characteristics of media and to accurately interpret information?

1-2 Information Ethics

Point!

1 Information Ethics

(1) ([1]**Information ethics**) : The foundational concepts and attitudes necessary for conducting appropriate activities in the information society.
This is regardless of the presence or absence of ([2]**laws**).

(2) Points to consider when disseminating information

[1] Characteristics of Information on the internet : Information is easily ([3]**spread**) and, once disseminated, it tends ([4]**not to disappear**) easily.

[2] Infringement of Others' Rights

- You must not publicly share images or videos for which others hold the ([5]**copyright**) without permission.
- You must not leak the ([6]**personal information**) of another person on the internet.
- You must not infringe upon others' ([7]**privacy**).
- You must not post comments that slander or defame others or engage in ([8]**online bullying**).

[3] ([9]**Geotag (Geotagging)**) : Information that includes latitude and longitude embedded in photos and videos taken with smartphones and mobile phones. There is a risk of being identified, such as having your home pinpointed based on the location where the photo was taken.

[4] ([10]**Disinformation and rumors**) : False information that is deliberately spread or baseless rumors.

2 Problems that occur with smartphones and social media

(1) ([11]**Social media**) : Services that provide platforms where individuals can connect with each other on the internet.

(2) Problems that occur with smartphones and social media

[1] ([12]**Internet addiction**) : A state where one becomes excessively immersed in the internet to the extent of interfering with daily life.

[2] ([13]**Using smartphone while walking**) : The act of using a smartphone while walking.

[3] ([14]**Cybercrime**) : The misuse of computers and networks for criminal activities.

[4] ([15]**Identity theft**) : The act of a third party pretending to be a certain individual or organization to steal IDs or passwords.

[5] ([16] **Leakage of personal information**) : When personal information that should remain confidential is disclosed to a third party.

Warm Up

Answer the following questions.

(1) What is the term for the foundational concepts and attitudes necessary for conducting appropriate activities in the information society?

(2) For the following statements **A** to **D**, mark "○" if the statement is correct, and "×" if it is incorrect.

A Social media is nothing more than a dangerous tool that causes trouble.

B There is a risk that your home address could be identified from geotags embedded in photos and videos.

C You can write whatever you want on an anonymous bulletin board because no one can ever know who wrote it.

D Be mindful that there are real people on the other side of the internet, and try to share information with consideration for their perspective.

Explanation

(1) Information ethics

(2) **A** Social media is a convenient tool on the internet that allows individuals to connect with each other. However, social media can be a dangerous tool if not used properly. Therefore, ×

B ○

C Never post defamatory content, disinformation, or rumors, even anonymously. Therefore, ×

D ○

Try

1 Answer the following questions.

(1) Fill in the blanks with the appropriate words.

[1] (　　) is the term for the foundational concepts and attitudes necessary for conducting appropriate activities in the information society.

[2] (　　) is the term for false information that is deliberately spread or baseless rumors.

[3] (　　) is the collective term for crimes involving the misuse of computers and networks.

(2) Choose the phrases that best fit into the blanks [1] to [3] from the options **A** to **D** below.

> When transmitting information on the internet, you must be very careful with handling others' ([1]) and with images or videos in which others have ([2]). Additionally, photos and videos taken with a smartphone have a feature that can add information such as the latitude and longitude of the location where they were taken, which is referred to as ([3]). ([3]) is convenient because it enables the confirmation or search of the location where a photo was taken, but caution is needed as there is a risk of having your home address identified.

A Copyright　**B** Personal Information　**C** Industrial Property Rights　**D** Geotag

(3) From the following options **A** to **D**, choose the one that is a general term for services that provide platforms for individuals to connect with each other on the internet. Answer by writing the corresponding letter.

A Social engineering　**B** Social marketing

C Social networking service　**D** Social network system

(4) The following statements **A** to **D** describe information ethics. Mark "○" if the statement is appropriate, and "×" if it is inappropriate.

A You can write freely on an anonymous bulletin board because no one can know who wrote it.

B Websites are legally required to always provide accurate information.

C You must be careful not only with your own personal information, but also with how you handle the personal information of others.

D For any content, it is better to use a bulletin board for information dissemination as it can reach a larger audience.

(5) Choose one statement that is appropriate as etiquette when using a smartphone from the options **A** to **D** below.

A I operated on a smartphone while driving my car.

B I turned off the smartphone at places like movie theaters or art museums.

C I posted negative comments about others on the internet.

D I sent my photo to a stranger I met online.

Exercise

1 Answer the following questions.

(1) Fill in the blanks with the appropriate words.

[1] (　　) is the term for a state in which a person prioritizes internet usage over aspects of life such as studies, work, physical and mental health, and is unable to control the time spent or the manner of internet usage.

[2] (　　) is the collective term for services that facilitate the building of relationships over the internet.

[3] (　　) is the term for location information that can be added to various media such as photos and videos.

(2) From the following options **A** to **H,** choose the phrases that best fit into the blanks [1] to [4].

(　[1]　) is the term for the foundational concepts and attitudes necessary for conducting appropriate activities in the information society. Even if something is not prohibited by (　[2]　), you should still refrain from engaging in inappropriate behavior. Furthermore, information on the internet is easily (　[3]　) and, once disseminated, it tends (　[4]　). With all of this in mind, we must be careful not to become involved in internet-related crimes or problematic behavior.

A	Information etiquette	**B**	Information ethics	**C**	Spread	**D**	Online backlash
E	To Disappear	**F**	Not To Disappear	**G**	Laws	**H**	System

(3) The following statements **A** to **D** describe information ethics. Mark "○" if the statement is appropriate, and "×" if it is inappropriate.

A When interacting through a network, it is important to communicate appropriately to avoid misunderstandings.

B You can write freely on an anonymous bulletin board because no one can ever know who wrote it.

C Websites are legally required to always provide accurate information.

D Since a celebrity was at a nearby shopping mall, I took a photo and shared it on social media. I assumed that this is OK since the photo was taken in a public place.

(4) Choose one correct statement regarding the use of the internet and social media from the options **A** to **D** below.

A It is acceptable to copy a commercially available CD and post it on social media without permission.

B Social media is nothing more than a dangerous tool that causes trouble.

C If you live a life disconnected from the internet, you won't be a victim of cybercrime.

D When disseminating information, ensure that it does not infringe on the privacy of others.